

1. Setup Grafana + Loki + Alloy

1.1 Deploy compose.yaml and alloy-config.alloy

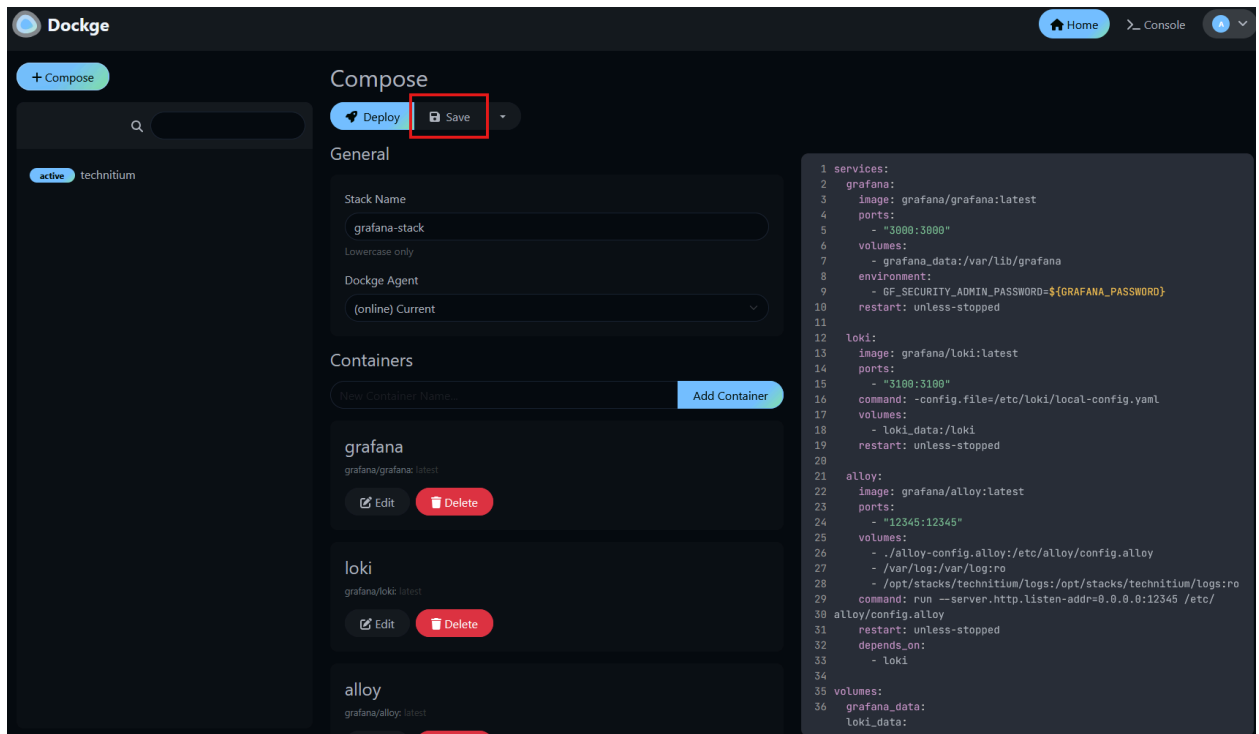
1. Copy the contents of the compose.yaml file here:
<https://github.com/davetjh/div0-homelab-workshop/blob/main/grafana-stack/compose.yaml>
2. Create a new stack in dockge and give it a name, e.g. grafana-stack
3. Paste the contents of the compose.yaml file.
4. Notice that there's an environment variable in the compose file called `GRAFANA_PASSWORD`. We will define this password in the `.env` section below the compose content. Put whatever password you want, but please remember it.

```
21 alloy:
22   image: grafana/alloy:latest
23   ports:
24     - "12345:12345"
25   volumes:
26     - ./alloy-config.alloy:/etc/alloy/config.alloy
27     - /var/log:/var/log:ro
28     - /opt/stacks/technitium/logs:/opt/stacks/technitium/logs:ro
29   command: run --server.http.listen-addr=0.0.0.0:12345 /etc/
30 alloy/config.alloy
31   restart: unless-stopped
32   depends_on:
33     - loki
34
35 volumes:
36   grafana_data:
37   loki_data:
```

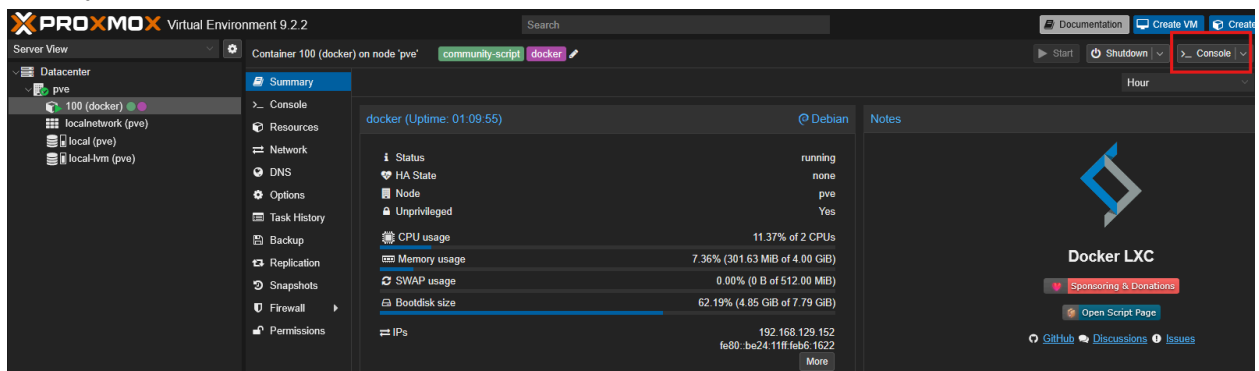
`.env`

```
1 GRAFANA_PASSWORD=P@ssw0rd123!|
```

- Click Save (instead of deploy). We will deploy it later



- Go back to the shell of the docker LXC. Refer to the screenshot below if you've closed it already.



- Run `cd /opt/stacks/grafana-stack` (or whatever name you've given the stack)
- Copy the contents of `alloy-config.alloy` from:
<https://github.com/davetjh/div0-homelab-workshop/blob/main/grafana-stack/alloy-config.alloy>
- In your shell, run `vi alloy-config.alloy`. Paste the content that you just copied in step 7 by pressing `Ctrl + Shift + v` together

10. Exit `vi` by typing `:wq` and pressing enter. Ensure that you have created `alloy-config.alloy` by running `cat alloy-config.alloy`

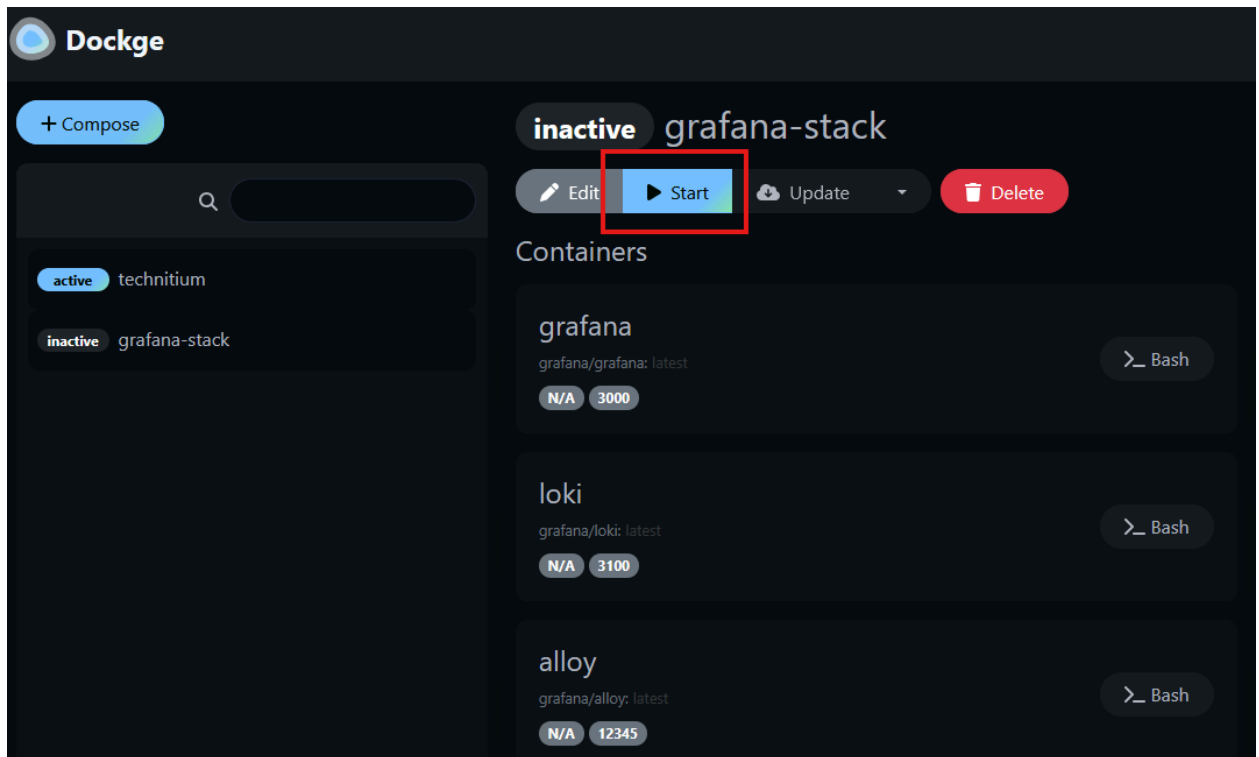
```
pve - Proxmox Console — Zen Browser
192.168.129.149:8006/?console=lxcl&xtermjs=1&vmid=100&vmname=docker&node=pve&cmd=

root@docker:~# cd /opt/stacks/grafana-stack
root@docker:/opt/stacks/grafana-stack# vi alloy-config.alloy
root@docker:/opt/stacks/grafana-stack# cat alloy-config.alloy
// Collect local log files
local.file_match "varlog" {
  path_targets = [{
    "__path__" = "/var/log/**/*.*log",
    "job"      = "varlog",
    "host"     = "technitium-lxc",
  }]
}

loki.source.file "varlog" {
  targets      = local.file_match.varlog.targets
  forward_to   = [loki.write.default.receiver]
}

local.file_match "technitium" {
  path_targets = [{
    "__path__" = "/opt/stacks/technitium/logs/*.*log",
    "job"      = "technitium",
    "host"     = "technitium-lxc",
  }]
}
```

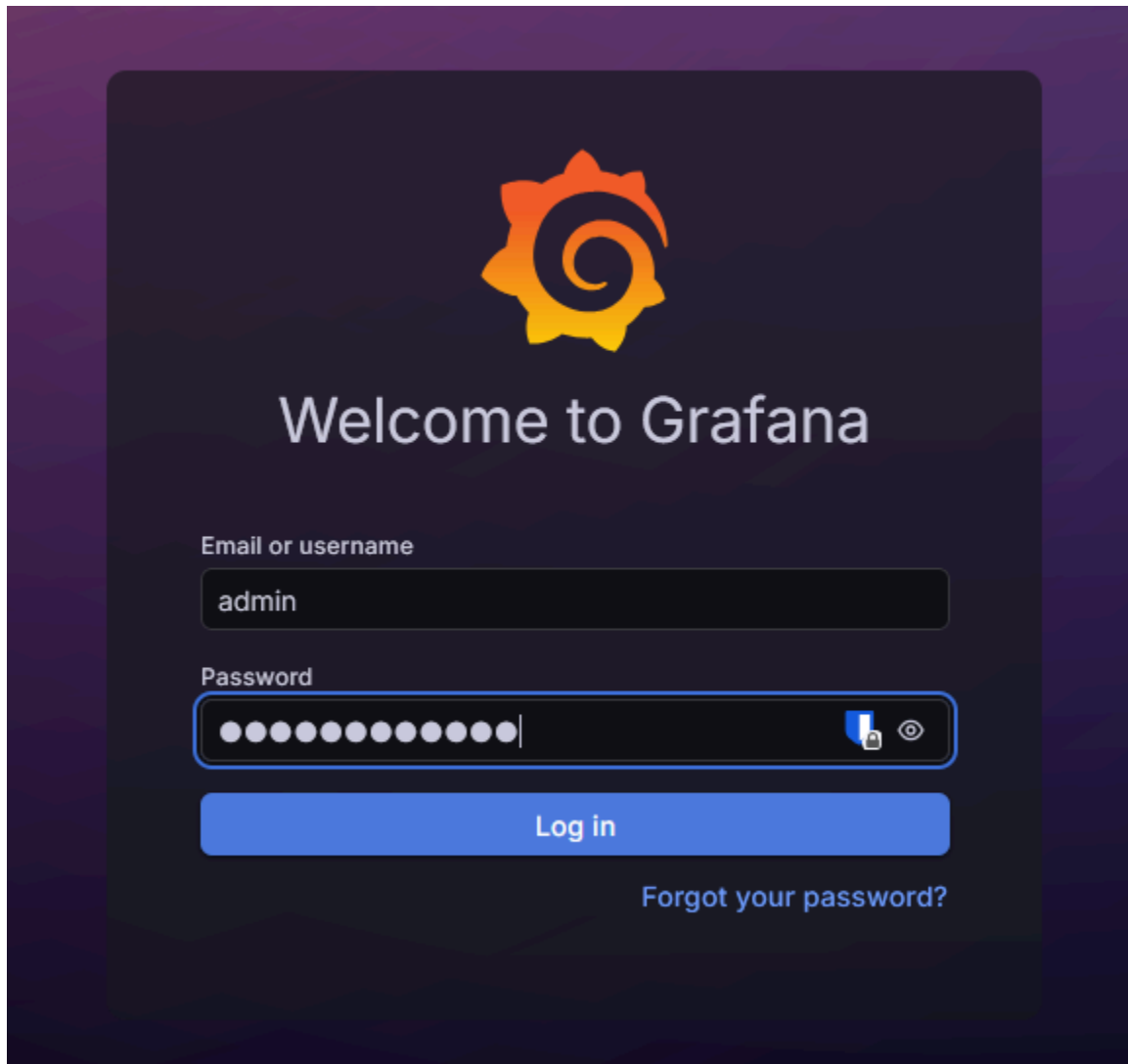
11. Deploy the stack in dockge.



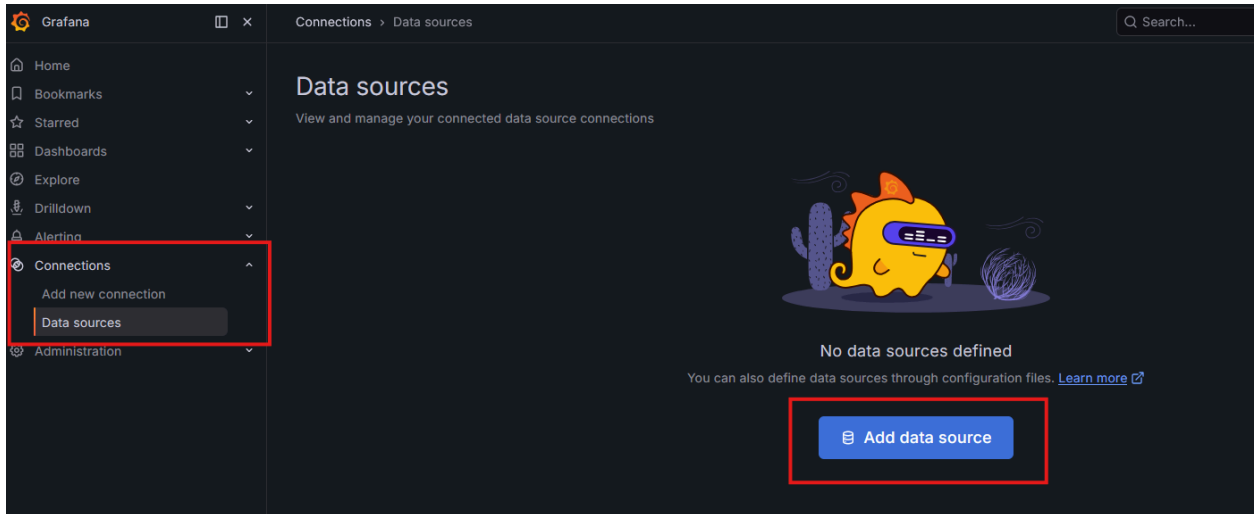
1.2 Access the Grafana dashboard and set Loki data source

1. Visit <https://<lxc-ip-addr>:3000>

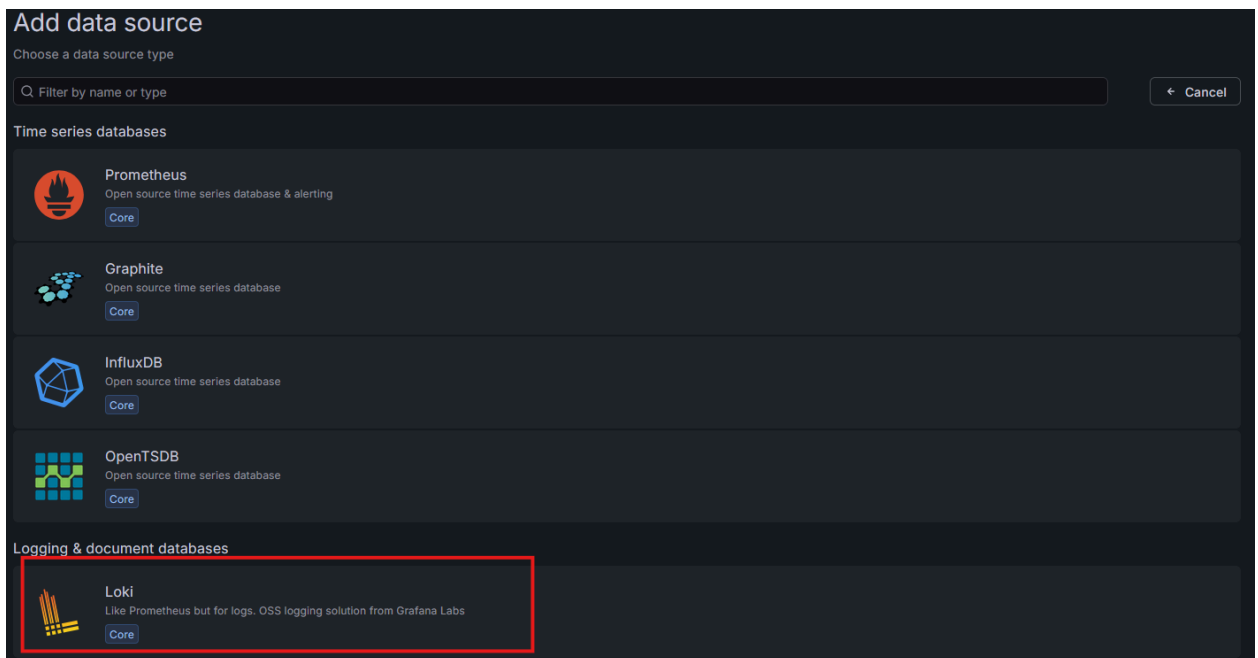
2. Login using the default credentials `admin:<your-chosen-password>`



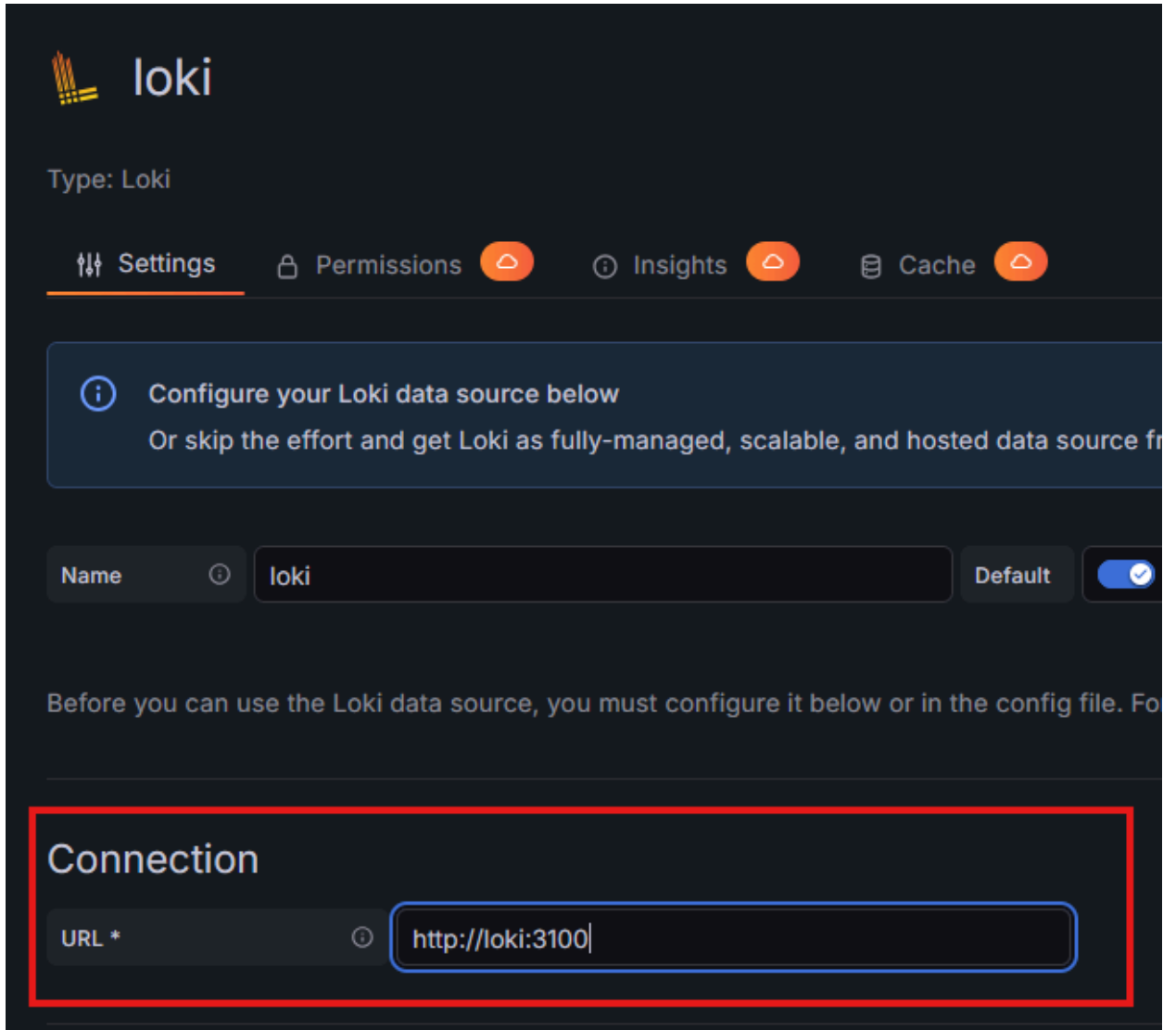
3. On the left menu bar, click on **Connections > Data Sources**. Then, click on Add data source.



4. Click on **Loki**

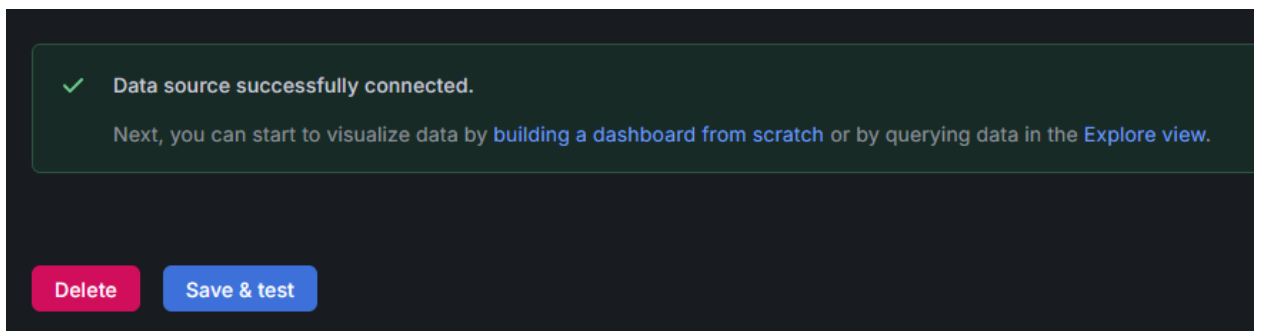


- Under Connection, key in `http://loki:3100`



The screenshot shows the 'loki' configuration page in a dark theme. At the top, there's a navigation bar with 'Settings', 'Permissions', 'Insights', and 'Cache'. Below this, a message says 'Configure your Loki data source below'. The 'Name' field is set to 'loki' and is marked as 'Default'. A red box highlights the 'Connection' section, which contains a 'URL *' field with the value 'http://loki:3100'.

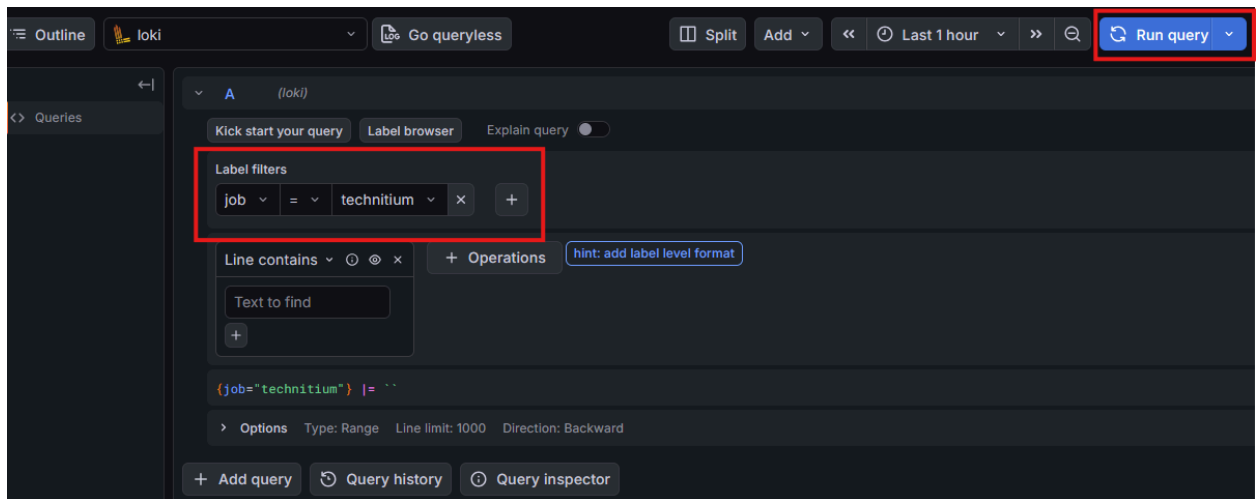
- Scroll down and click **Save & test**. If configured well, the data source will be successfully connected.



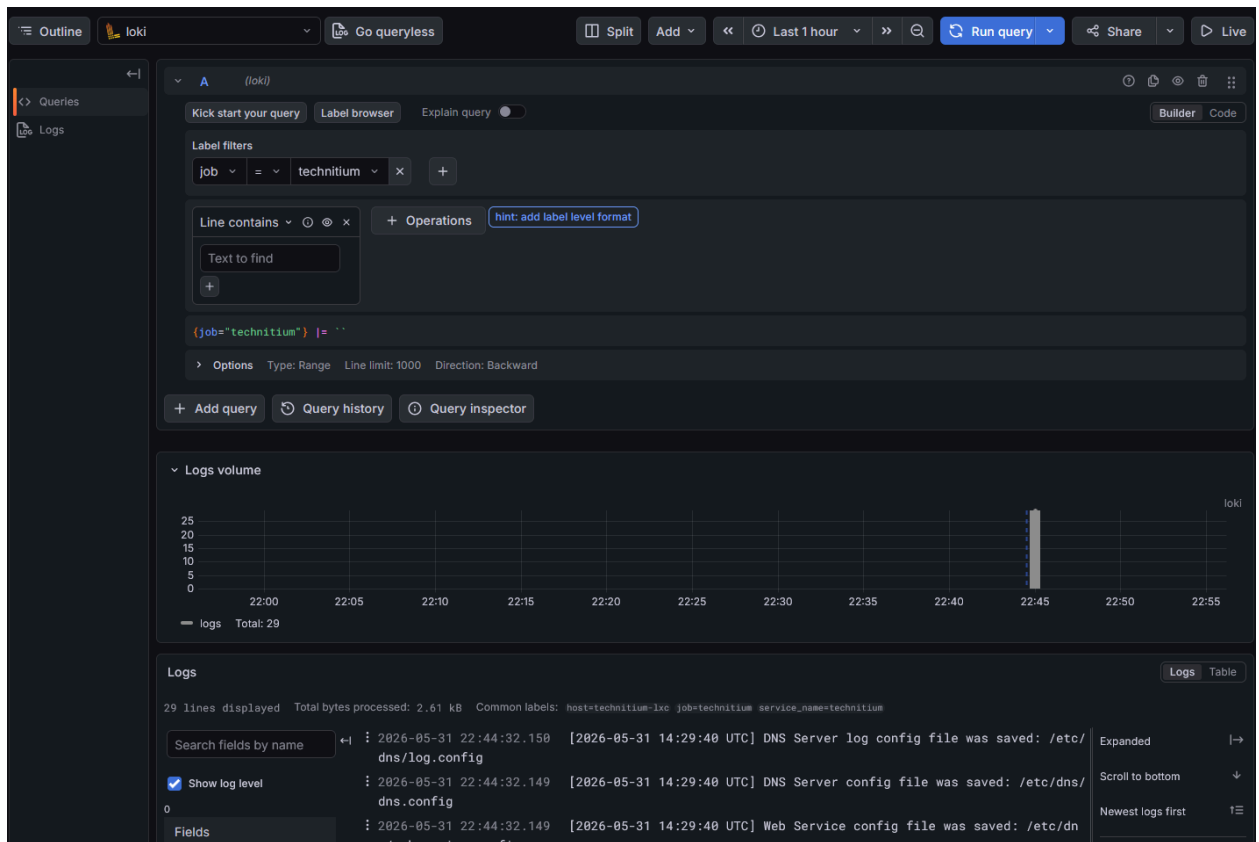
The screenshot shows a success message in a green box: 'Data source successfully connected.' Below this, it says 'Next, you can start to visualize data by building a dashboard from scratch or by querying data in the Explore view.' At the bottom, there are two buttons: 'Delete' and 'Save & test'.

2. Query Logs

1. In the left menu bar, click on **Explore**
2. Select the following:
 - Select label: **job**
 - Select value: **technitium**
3. Click on Run Query.



4. You should be able to see some logs below.



5. Now try making some DNS queries through the Technitium DNS server. For demonstration purposes, use **nslookup**.

nslookup google.com <technitium-lxc-ip-addr>

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Jun Hong Tan> nslookup google.com 192.168.129.152
Server: UnKnown
Address: 192.168.129.152

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4003:c00::66
          2404:6800:4003:c00::8b
          2404:6800:4003:c00::64
          2404:6800:4003:c00::8a
          142.250.4.139
          142.250.4.100
          142.250.4.101
          142.250.4.138
          142.250.4.102
          142.250.4.113
```

6. Now run the query in grafana again. You should see the DNS query and response in the logs.

```
Logs
38 lines displayed Total bytes processed: 3.78 kB Common labels: host=technitium-lxc job=technitium service_name=technitium

↳ : 2026-05-31 22:59:20.528 [2026-05-31 14:59:20 UTC] [192.168.129.1:56703] [UDP] QNAME: google.com; QTYPE: AAAA; QCLASS: IN; RCODE: NoError; ANSWER: [2404:6800:4003:c00::66, 2404:6800:4003:c00::8b, 2404:6800:4003:c00::64, 2404:6800:4003:c00::8a]
: 2026-05-31 22:59:20.528 [2026-05-31 14:59:20 UTC] [192.168.129.1:56696] [UDP] QNAME: google.com; QTYPE: A; QCLASS: IN; RCODE: NoError; ANSWER: [142.250.4.139, 142.250.4.100, 142.250.4.101, 142.250.4.138, 142.250.4.102, 142.250.4.113]
```

Congratulations! You've deployed your very own log monitoring solution for DNS logs.